

The Freshman Survival Cheat Sheet Bundle

Everything you wish someone told you before Day 1

BIOLOGY 101 — CORE CONCEPTS CHEAT SHEET

THE CELL

Prokaryotes: No nucleus (bacteria, archaea) | Eukaryotes: Have nucleus (plants, animals, fungi)

!Nucleus ! stores DNA, controls cell activity

!Mitochondria ! ATP production ("powerhouse"), has own DNA

!Ribosomes ! protein synthesis (found on rough ER or free in cytoplasm)

!Endoplasmic Reticulum (ER) ! Rough ER: proteins | Smooth ER: lipids & detox

!Golgi Apparatus ! packages/ships proteins (like FedEx for the cell)

!Lysosomes ! digest waste (only in animal cells)

!Cell Wall ! rigid outer layer (plants, fungi, bacteria — NOT animal cells)

!Chloroplasts ! photosynthesis (plants only), has own DNA

DNA & GENETICS (HIGH PRIORITY — always on exams)

!DNA ! RNA ! Protein (Central Dogma — memorize this direction)

!Transcription: DNA ! mRNA (happens in nucleus)

!Translation: mRNA ! protein (happens at ribosome)

!Codons: 3-base sequences on mRNA that code for amino acids

!Anticodon: complementary sequence on tRNA

!Mutation types: Substitution (one base changes) | Insertion/Deletion (frameshift — much worse)

!Dominant allele (A) masks recessive (a) | Homozygous (AA or aa) | Heterozygous (Aa)

!Punnett Square: tool for predicting offspring genotype ratios

!Mitosis: cell division for GROWTH (2 identical daughter cells)

!Meiosis: cell division for SEX CELLS (4 cells, half the chromosomes)

EVOLUTION & ECOLOGY

!Natural Selection: variation ! selection pressure ! differential reproduction ! evolution

!Darwin's 4 postulates: variation exists, variation is heritable, more offspring than survive, variation affects survival

!Hardy-Weinberg: $p^2 + 2pq + q^2 = 1$ ($p + q = 1$) — used for population genetics problems

!Food chains: Producer ! Primary Consumer ! Secondary Consumer ! Tertiary Consumer

!Energy transfer: ~10% of energy moves up each trophic level

!Symbiosis types: Mutualism (+/+) | Commensalism (+/0) | Parasitism (+/-)

CALCULUS I — FORMULA SHEET

LIMITS

! $\lim_{x \rightarrow c} f(x) = L$ means $f(x)$ approaches L as x approaches c

! Direct substitution: plug in the value first — if you get $0/0$, try factoring

! L'Hôpital's Rule: if $0/0$ or ∞/∞ , take derivative of top and bottom separately

! Continuity: f is continuous at c if $\lim_{x \rightarrow c} f(x) = f(c)$

DERIVATIVES — RULES TO MEMORIZE

! Power Rule: $d/dx [x^n] = nx^{n-1}$

! Constant Rule: $d/dx [c] = 0$

! Product Rule: $d/dx [fg] = f'g + fg'$

! Quotient Rule: $d/dx [f/g] = (f'g - fg') / g^2$

! Chain Rule: $d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$

! $d/dx [\sin x] = \cos x$ | $d/dx [\cos x] = -\sin x$

! $d/dx [\tan x] = \sec^2 x$ | $d/dx [e^x] = e^x$

! $d/dx [\ln x] = 1/x$ | $d/dx [a^x] = a^x \ln(a)$

! Implicit differentiation: differentiate both sides, treat y as a function of x

APPLICATIONS OF DERIVATIVES

! Critical points: set $f'(x) = 0$, solve for x

! First Derivative Test: f' changes $+$ to $-$! local max | $-$ to $+$! local min

! Second Derivative Test: $f''(c) < 0$! local max | $f''(c) > 0$! local min

! Inflection point: where f'' changes sign

! Related rates: differentiate both sides with respect to time (t)

! Optimization: find critical points, check endpoints on closed interval

INTEGRALS

! $\int x^n dx = x^{n+1}/(n+1) + C$ ($n \neq -1$)

! $\int 1/x dx = \ln|x| + C$

! $\int e^x dx = e^x + C$ | $\int \sin x dx = -\cos x + C$ | $\int \cos x dx = \sin x + C$

! Fundamental Theorem Part 1: $d/dx \int_a^x f(t) dt = f(x)$

! Fundamental Theorem Part 2: $\int_a^b f(x) dx = F(b) - F(a)$

! U-substitution: let u = inner function, du = derivative $\cdot dx$

! Area between curves: $\int_a^b [\text{top} - \text{bottom}] dx$

INTRO PSYCHOLOGY — KEY THEORIES & RESEARCHERS

MAJOR PERSPECTIVES (know all 7)

- !Biological: behavior explained by brain, genes, neurotransmitters
- !Psychodynamic (Freud): unconscious mind, childhood experiences, id/ego/superego
- !Behavioral (Watson, Skinner): behavior shaped by environment and conditioning
- !Cognitive: mental processes — memory, perception, problem-solving, decision-making
- !Humanistic (Maslow, Rogers): free will, self-actualization, human potential
- !Sociocultural: influence of society, culture, and relationships on behavior
- !Evolutionary: behavior evolved because it helped ancestors survive/reproduce

HIGH-PRIORITY RESEARCHERS & STUDIES

- !Pavlov ! Classical conditioning (dogs + bell + food)
- !Skinner ! Operant conditioning (reinforcement schedules, Skinner box)
- !Milgram ! Obedience study (65% shocked to max voltage)
- !Zimbardo ! Stanford Prison Experiment (roles affect behavior)
- !Bandura ! Bobo doll (observational learning/social learning theory)
- !Maslow ! Hierarchy of Needs (physiological ! safety ! love ! esteem ! self-actualization)
- !Kohlberg ! Moral development (preconventional ! conventional ! postconventional)
- !Piaget ! Cognitive development (sensorimotor ! preoperational ! concrete ! formal)
- !Erikson ! Psychosocial stages (8 stages across the lifespan)
- !Broca ! Language production area in left frontal lobe
- !Wernicke ! Language comprehension area in left temporal lobe

MEMORY & LEARNING (almost always on exams)

- !Encoding ! Storage ! Retrieval (three stages of memory)
- !Sensory Memory ! Short-Term (7±2 items) ! Long-Term
- !Working memory: active processing (like mental RAM)
- !Explicit memory: conscious recall (episodic + semantic)
- !Implicit memory: unconscious (procedural skills, priming)
- !Proactive interference: old info disrupts new | Retroactive: new disrupts old
- !Spacing effect: distributed practice beats cramming (use this!)
- !Classical conditioning: neutral stimulus + unconditioned stimulus ! conditioned response
- !Operant conditioning: behavior followed by consequence (reinforcement or punishment)
- !Continuous reinforcement: faster to learn | Variable ratio: hardest to extinguish

MACROECONOMICS — CORE CONCEPTS

KEY MEASUREMENTS

- !GDP = $C + I + G + (X - M)$ [Consumption + Investment + Government + Net Exports]
- !Real GDP: adjusted for inflation | Nominal GDP: current prices
- !GDP per capita = $GDP \div Population$ (measures standard of living)
- !CPI (Consumer Price Index): measures inflation using a basket of goods
- !Inflation rate = $(CPI_t - CPI_{t-1}) / CPI_{t-1} \times 100$
- !Unemployment rate = $(Unemployed / Labor\ Force) \times 100$
- !Types: Frictional (between jobs), Structural (skills mismatch), Cyclical (recession-caused)
- !Natural rate of unemployment: frictional + structural (never zero)

SUPPLY & DEMAND

- !Law of Demand: price up ! quantity demanded down (inverse)
- !Law of Supply: price up ! quantity supplied up (direct)
- !Equilibrium: where supply curve meets demand curve
- !Demand shifters: income, prices of related goods, tastes, expectations, number of buyers
- !Supply shifters: input costs, technology, expectations, number of sellers
- !Price ceiling (below equilibrium) ! shortage | Price floor (above equilibrium) ! surplus
- !Elasticity: how sensitive quantity is to price change
- !Elastic demand (>1): luxury goods | Inelastic demand (<1): necessities

MONETARY & FISCAL POLICY

- !Fiscal Policy (government): spending + taxes | Expansionary = spend more/tax less
- !Monetary Policy (Federal Reserve): controls money supply and interest rates
- !Contractionary policy: raise rates, reduce money supply ! fight inflation
- !Expansionary policy: lower rates, increase money supply ! fight recession
- !Federal Funds Rate: interest rate banks charge each other overnight
- !Multiplier effect: initial spending ! more spending throughout economy
- !Quantity Theory of Money: $MV = PQ$ (Money $\times$ Velocity = Price Level $\times$ Output)
- !Phillips Curve: inverse relationship between inflation and unemployment (short run)

ENGLISH COMPOSITION — ESSAY FRAMEWORK

THE FIVE-PARAGRAPH STRUCTURE (foundation)

!Paragraph 1 — Introduction: Hook ! Background ! Thesis

!Paragraphs 2-4 — Body: Topic sentence ! Evidence ! Analysis ! Transition

!Paragraph 5 — Conclusion: Restate thesis (differently) ! Synthesize points ! So what?

Strong thesis formula: [Specific claim] because [reason 1], [reason 2], and [reason 3].

Weak: "Social media has effects on students." ! Strong: "Social media use above 3 hours daily decreases college student GPA by damaging sleep quality, reducing focus, and creating social anxiety."

THE PARAGRAPH STRUCTURE (PEEL or TIQA)

!T — Topic sentence: state the point of this paragraph

!I — Introduce evidence: signal phrase + quote or paraphrase

!Q — Quote/evidence: "According to [Author], ..."

!A — Analysis: explain what the evidence proves (this is where most students lose points)

Analysis sentence starters: "This demonstrates...", "This suggests...", "This is significant because...", "In other words...", "This evidence supports the claim that..."

COMMON MISTAKES THAT COST YOU GRADES

!Thesis too vague — must take a specific, arguable position

!Quoting without analyzing — evidence alone earns no points; explain what it proves

!Starting a paragraph with a quote — always introduce it first

!Using "I believe" or "In my opinion" in academic essays — just state the claim

!Weak conclusions that just summarize — connect to broader significance

!Not citing within the paper — every claim needs a source in-text

COLLEGE LIFE TIPS NOBODY TELLS YOU

!Go to office hours in week 1, before you need them. Professors remember faces.

!Email professors with your name + section. They grade names they recognize higher.

!Get the syllabus on day 1 and enter all due dates into your phone immediately.

!Sleep is your #1 study tool. 7 hours > 5 hours of studying on 4 hours of sleep.

!Sit in the first three rows of every class. You will learn 30% more.

!Use Rate My Professor before registration, but take it with a grain of salt.

!Free stuff: student discounts, campus events, health center, tutoring center — use all of them.

!Drink water and eat breakfast before exams. It actually affects performance measurably.

!Your campus career center is criminally underused. Go in freshman year, not senior year.

!Join one club the first week. You will procrastinate if you wait until you "settle in."

!Your RA is a resource, not just a rule enforcer. They know things.

!The library has private study rooms — reserve them. Free and productive.

!Set a hard deadline to start papers 3 days before they're due, not the night before.

!Learn your professor's grading rubric. It tells you exactly what to write.

!Network with upperclassmen in your major. They have the notes, tips, and connections.

